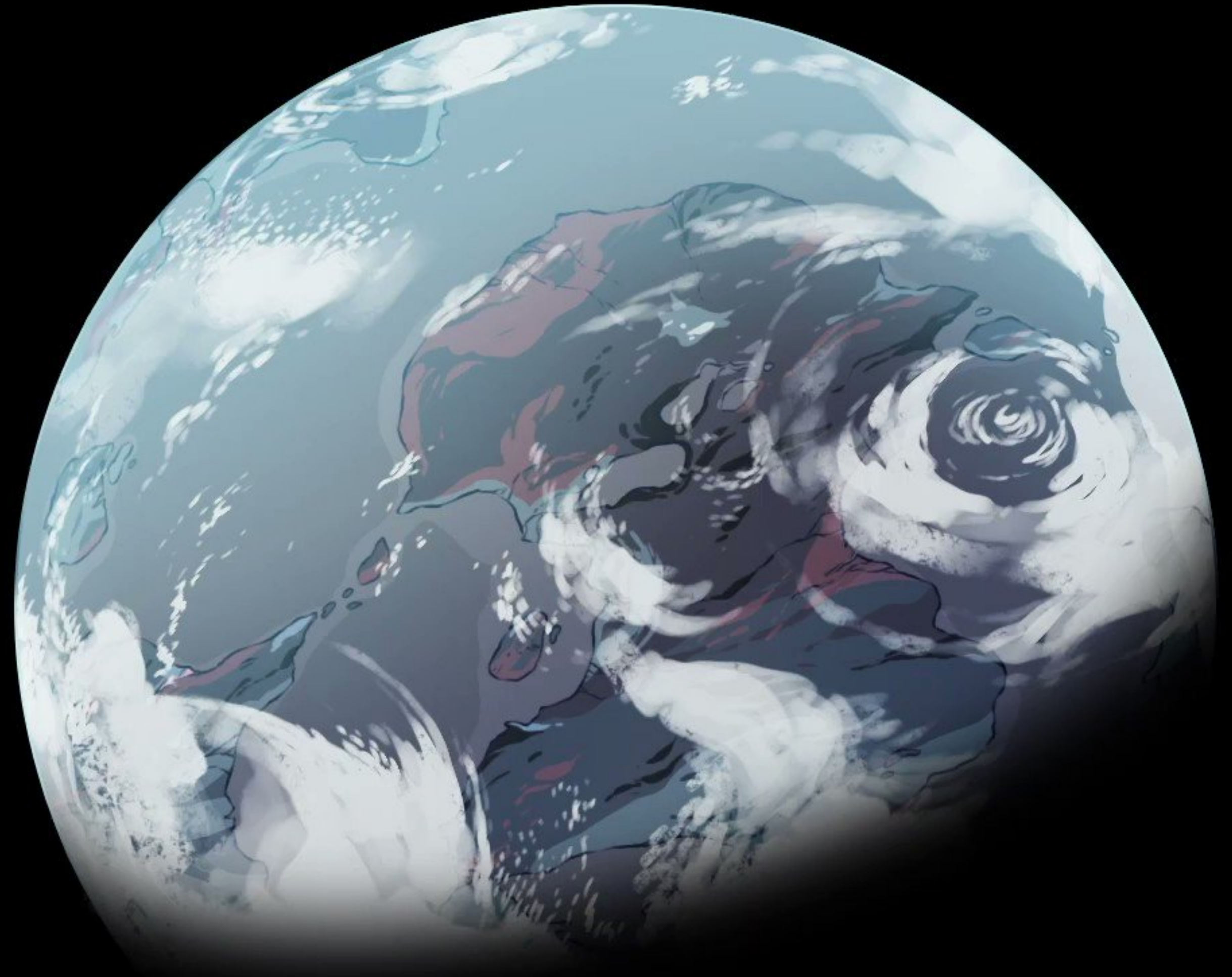


Science & Science Fiction

33-120 | Prof. Katie Breivik



Announcements

- ▶ Midterm Film Critique due today @ midnight
- ▶ Problem 4 due Wednesday @ midnight
- ▶ Exploration paper 2 due Friday @ midnight

Today: Forces, Energy & Power

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What is the difference between
a force, energy and power?



Star Wars

Directed:
George Lucas

20th Century Fox
(1977)



“The Force is.. an
energy field...”
- Obi-Wan Kenobi



“The Force is.. an
energy field...”

- Obi-Wan Kenobi

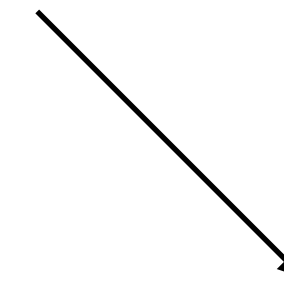
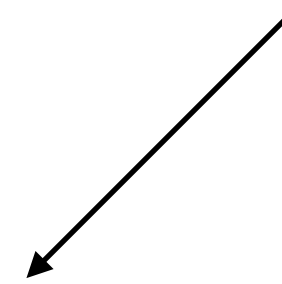
“A Force is.. the
gradient of an
energy field...”

- Katie Breivik

(and all Physics professors)

“A Force is.. the *gradient of* an energy field...”

What is an **energy field**?



What is a field?

A field is a physical quantity
that has a value for any point
in space and time

What is energy?

A quantitative property that is
transferred to a physical
system in the form of “work”,
heat, or light

Example: weight on the end of a spring



Potential energy in a stretched spring: $U = \frac{1}{2}kx^2$

Force exerted by a stretched spring: $|F| = \nabla \left(\frac{1}{2}kx^2 \right)$
 $= kx$

What sign should the force have?

$$F = -\nabla U$$

Example: weight on the end of a spring



Potential energy in a stretched spring: $U = \frac{1}{2}kx^2$

Force exerted by a stretched spring: $F = -kx$

Kinetic energy: the energy that an object
possesses due to its motion

In classical mechanics: $K = \frac{1}{2}mv^2$

Example: weight on the end of a spring



Potential energy in a stretched spring: $U = \frac{1}{2}kx^2$

Force exerted by a stretched spring: $F = -kx$

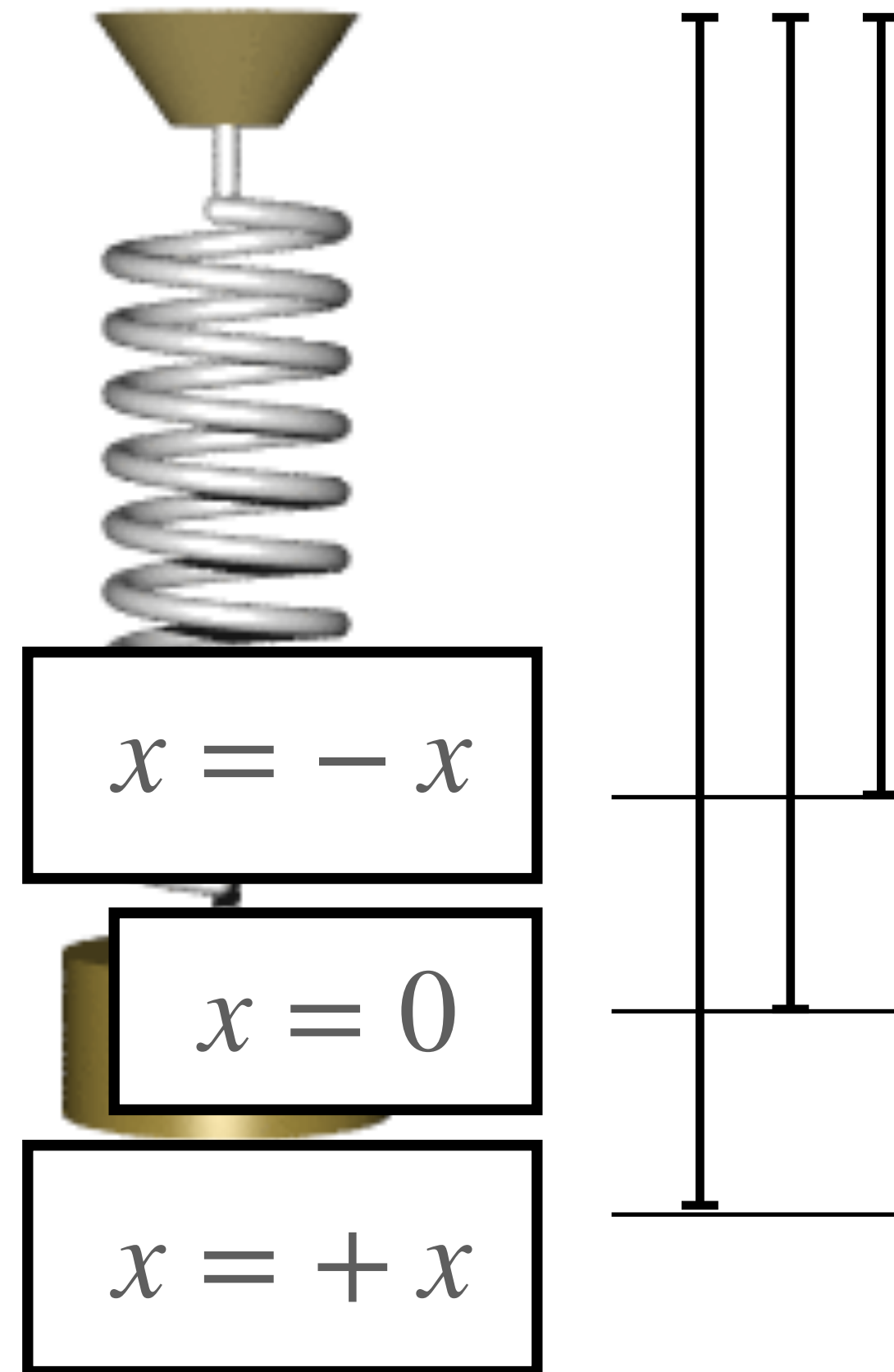
Kinetic energy of the weight: $K = \frac{1}{2}mv^2$

The total energy is: $E = U + K$

Example: weight on the end of a spring

The total energy *at any time* is:

$$E = U + K$$



$$E_{\text{bottom}} = ?$$

$$v_{\text{bottom}} = 0$$

$$K_{\text{bottom}} = 0$$

$$U_{\text{bottom}} = \frac{1}{2}kx^2$$

$$E_{\text{middle}} = ?$$

$$v_{\text{middle}} = v$$

$$K_{\text{middle}} = \frac{1}{2}mv^2$$

$$U_{\text{middle}} = 0$$

$$E_{\text{top}} = ?$$

$$v_{\text{top}} = 0$$

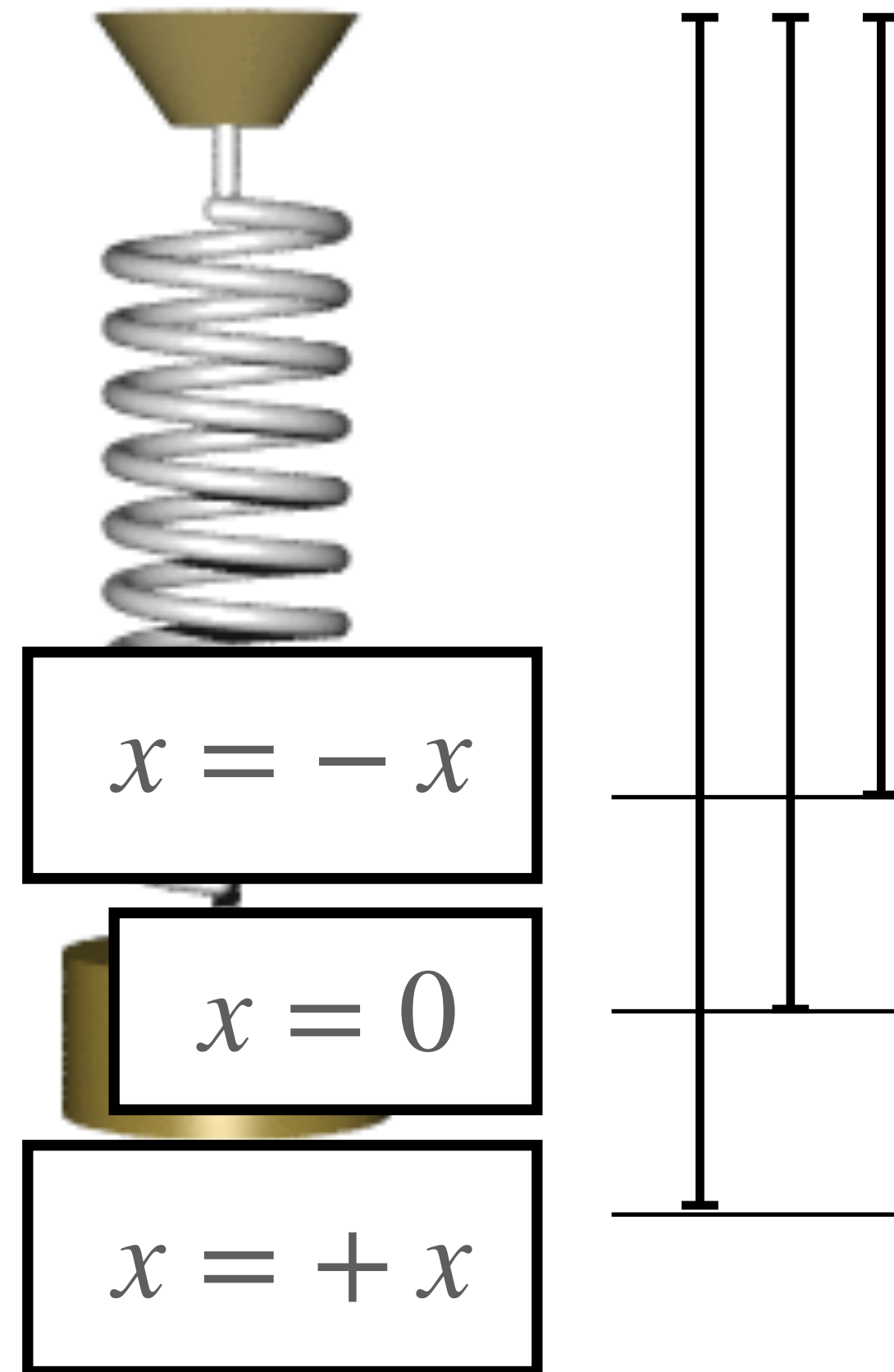
$$K_{\text{top}} = 0$$

$$U_{\text{top}} = \frac{1}{2}kx^2$$

Example: weight on the end of a spring

The total energy *at any time* is:

$$E = U + K$$



$$E_{\text{bottom}} = ?$$

$$E_{\text{middle}} = ?$$

$$E_{\text{top}} = ?$$

$$K_{\text{bottom}} = 0$$

$$K_{\text{middle}} = \frac{1}{2}mv^2$$

$$K_{\text{top}} = 0$$

$$U_{\text{bottom}} = \frac{1}{2}kx^2$$

$$U_{\text{middle}} = 0$$

$$U_{\text{top}} = \frac{1}{2}kx^2$$

Conservation of energy: $\Delta E = 0 \longrightarrow \frac{1}{2}kx^2 = \frac{1}{2}mv^2$



Spider Man 2

Directed:
Sam Raimi

Columbia Pictures
(2004)

How realistic is this situation?

Is Spiderman's facial expression warranted here?



An assumption: the spider web is *basically* a fancy spring

If the train is stopped, what does the energy budget look like?

$$\begin{aligned} E_{\text{final}} &= U + K = \frac{1}{2}kx^2 + \frac{1}{2}mv^2 \\ &= \frac{1}{2}kx^2 + \frac{1}{2}m(0) = \frac{1}{2}kx^2 \end{aligned}$$

An assumption: the spider web is *basically* a fancy spring

If the train is stopped, what does the energy budget look like?

$$E_{\text{final}} = U + \overset{0}{\cancel{K}} = \frac{1}{2}kx^2$$

Before the web is deployed, what does the energy budget look like?

$$E_{\text{initial}} = U + K = \frac{1}{2}kx^2 + \frac{1}{2}mv^2$$

An assumption: the spider web is *basically* a fancy spring

If the train is stopped, what does the energy budget look like?

$$E_{\text{final}} = U + \overset{0}{\cancel{K}} = \frac{1}{2}kx^2$$

Before the web is deployed, what does the energy budget look like?

$$E_{\text{initial}} = U + K = \frac{1}{2}kx^2 + \frac{1}{2}mv^2 = \frac{1}{2}k(0)^2 + \frac{1}{2}mv^2$$

An assumption: the spider web is *basically* a fancy spring

If the train is stopped, what does the energy budget look like?

$$E_{\text{final}} = U + \cancel{K}^0 = \frac{1}{2}kx^2$$

Before the web is deployed, what does the energy budget look like?

$$E_{\text{initial}} = \cancel{U}^0 + K = \frac{1}{2}mv^2$$

An assumption: the spider web is *basically* a fancy spring

Another assumption: energy is conserved

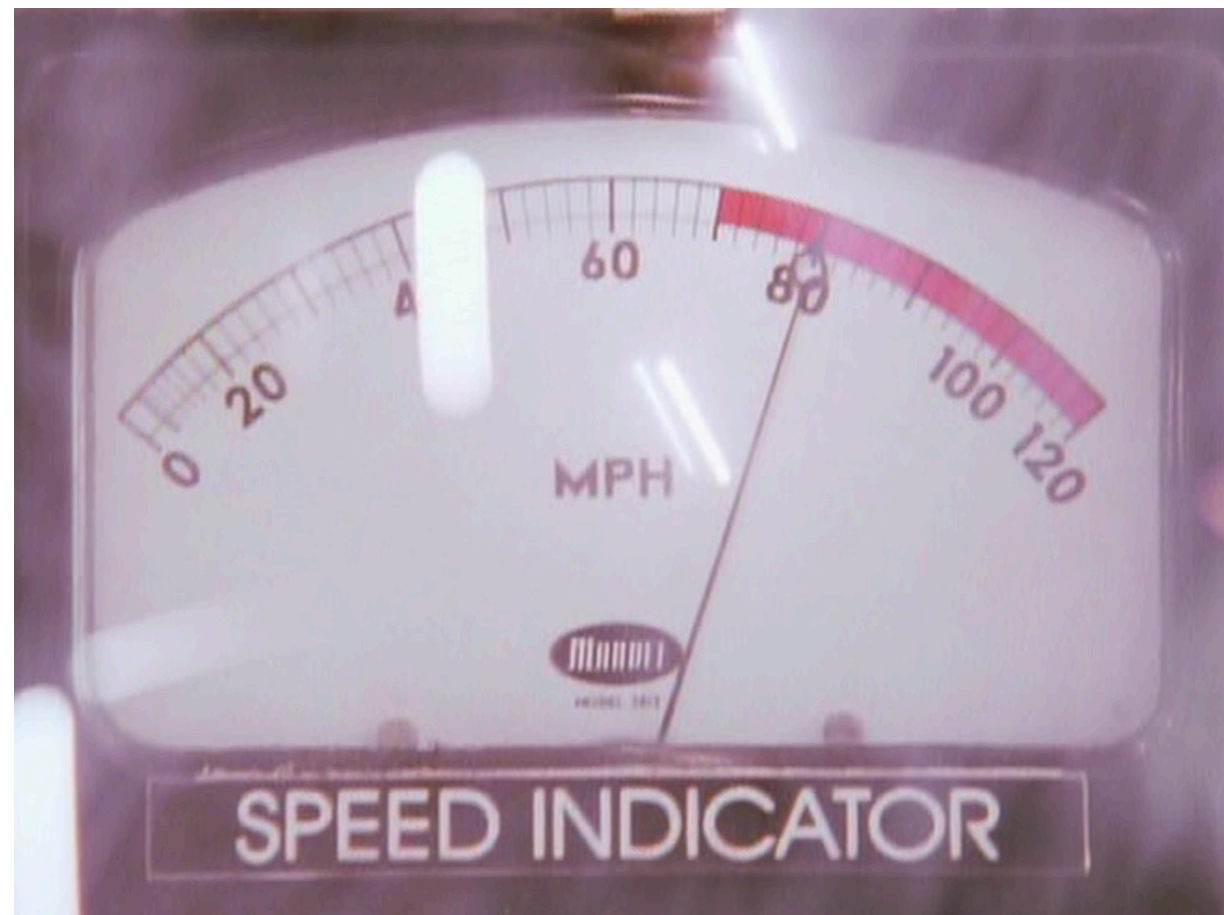
$$\Delta E = E_{\text{final}} - E_{\text{initial}} = 0 \quad \longrightarrow \quad E_{\text{final}} = E_{\text{initial}}$$

$$\frac{1}{2}kx^2 = \frac{1}{2}mv^2 \quad \longrightarrow \quad k = m\frac{v^2}{x^2}$$

We need some estimates for v, x, m !

We need some estimates for $v, x, m!$

$$k = m \frac{v^2}{x^2}$$



$v = 80 \text{ mph} = 35.8 \text{ m/s}$ Guess 1 NYC avenue: $x \simeq 230 \text{ m}$

Look up the mass of an mta car: $m = 85,000 \text{ lb} = 35,555 \text{ kg}$

Multiply by 8 cars per train: $m_{\text{train}} = 284,440 \text{ kg}$

Calculate the force constant for the web:

$$k = m \frac{v^2}{x^2} = 284,440 \text{ kg} \times \frac{(35.8 \text{ m/s})^2}{(230)^2} = 6890 \text{ N/m}$$

So what force does Spiderman feel?

$$|F| = kx = 6890 \text{ N/m} \times 230\text{m} = 1.6 \text{ million N}$$

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$$|F| = kx = 6890 \text{ N/m} \times 230 \text{ m} = 1.6 \text{ million N} = 170 \text{ tons}$$

Is this plausible??

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$$|F| = kx = 1.6 \text{ million N} = 170 \text{ tons}$$

Young's modulus: $Y = \frac{\text{stress}}{\text{strain}} = \frac{\text{Force / area}}{\Delta L / L}$

Guess: area = 1 cm² & $L = 20 \text{ ft} = 6 \text{ m}$ & note: $x = \Delta L$

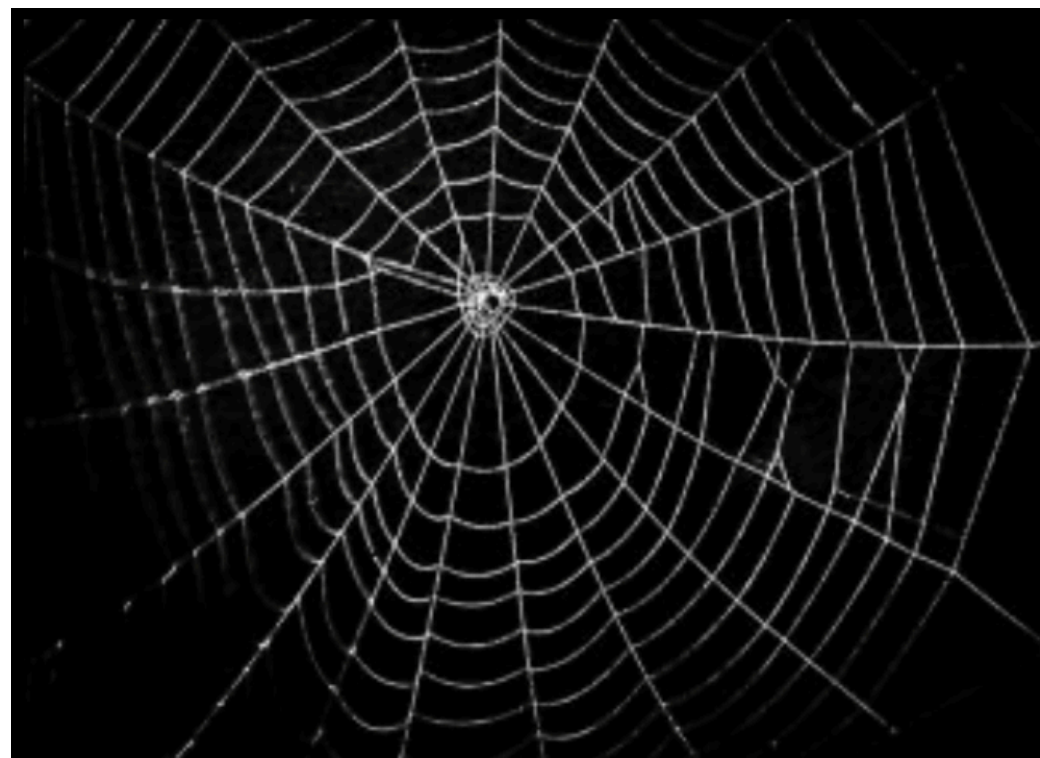
$$Y = \frac{F/A}{\Delta L/L} = \frac{kx/A}{x/L} = kL/A$$

Is this plausible??

$$Y = \frac{F/A}{\Delta L/L} = \frac{kx/A}{x/L} = kL/A$$

$$Y = kL/A = 6890 \text{ N/m} \times 6 \text{ m} / 0.0001 \text{ m}^2 = 413 \times 10^6 \text{ N/m}^2$$

Spiderman web: $Y = 413 \text{ MPa}$



Radial spider silk: 1.5 GPa

Spiral spider silk: 5-10 MPa

Omg it is!

$$\text{Power} = \frac{\text{Energy}}{\text{time}}$$

Power is the rate at which energy is 'used' in a system

Chemical energy: calories vs Calories

$$1 \text{ calorie} = 4.1868 \text{ J}$$

Heat energy required
to raise temperature
of 1 gram of water by
 1°C

$$1 \text{ Calorie} = 1,000 \text{ calories}$$

$$1 \text{ Calorie} = 4186.8 \text{ J}$$

These are both forms of **chemical energy**

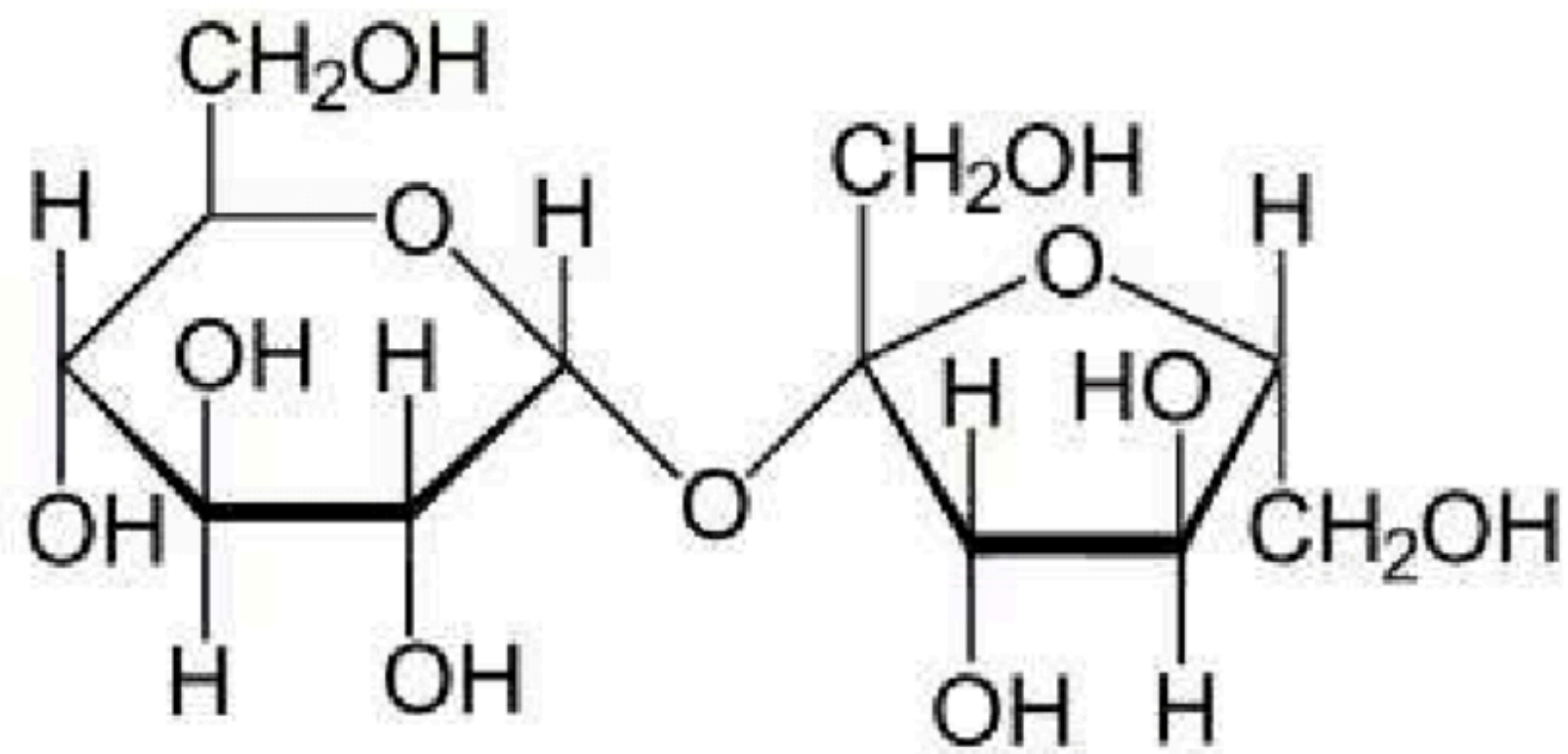


Instant oatmeal

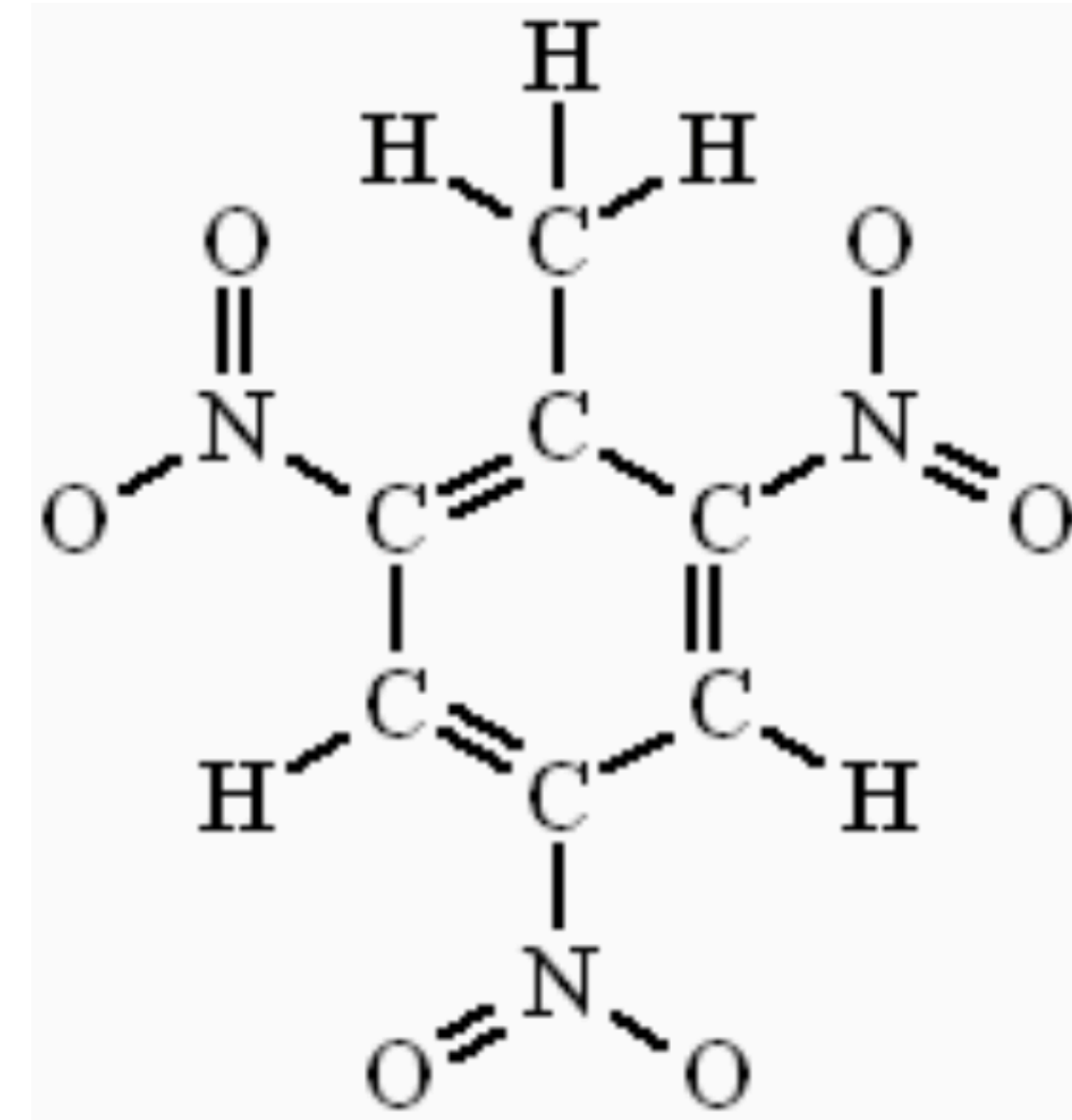


TNT: trinitrotoluene

These are both forms of **chemical energy**



Instant oatmeal
(sugar)



TNT: trinitrotoluene

These are both forms of **chemical energy**



Instant oatmeal

3.8 Calories / gram



TNT: trinitrotoluene

1 Calorie / gram

These are both forms of **chemical energy**



Instant oatmeal
1200 Calories per box



TNT: trinitrotoluene
1000 Calories per kg

These are both forms of **chemical energy**



What is the deal here??

$$\text{Power} = \text{energy} / \text{time}$$



Instant oatmeal

1200 Calories per box

TNT: trinitrotoluene

1000 Calories per kg

These are both forms of **chemical energy**



$$\text{Power} = 1000 \text{ Calories} / 0.1 \text{ seconds}$$



$$\text{Power} = 1200 \text{ Calories} / 1 \text{ week}$$

Instant oatmeal
1200 Calories per box

TNT: trinitrotoluene
1000 Calories per kg

Reminders!

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Next time: Efficient use of energy